

Dataset Overview

1. Purpose

This document describes the datasets and dataset-like assets currently stored in the repository. It is meant to be the single place to check what each folder is for, which files are meant for training or testing, and which files are only for demos or interface checks.

2. Dataset Groups

Group	Location	Main contents
Demo dataset	[`data/demo/`](/d:/Project/Dyslexia_Detection_System/data/demo)	manifest CSVs, audio samples,
Collection dataset	[`data/collection/`](/d:/Project/Dyslexia_Detection_System/data/collection)	raw collection workspace, pro
Collection test dataset	[`data/collection_test/`](/d:/Project/Dyslexia_Detection_System/data/collection_test)	test collection workspace and
Benchmark datasets	[`data/benchmarks/`](/d:/Project/Dyslexia_Detection_System/data/benchmarks)	tough manifests and hard fam
Biomarker test CSVs	[`data/biomarker_tests/`](/d:/Project/Dyslexia_Detection_System/data/biomarker_tests)	synthetic biomarker CSVs for E

3. Demo Dataset Details

The demo dataset is the quickest way to run the project locally.

3.1 Main files

File	Role
[`data/demo/manifest.csv`](/d:/Project/Dyslexia_Detection_System/data/demo/manifest.csv)	Small starte
[`data/demo/manifest_anonymized.csv`](/d:/Project/Dyslexia_Detection_System/data/demo/manifest_anonymized.csv)	Anonymize
[`data/demo/augmented_manifest.csv`](/d:/Project/Dyslexia_Detection_System/data/demo/augmented_manifest.csv)	Augmented
[`data/demo/audio_augmented_manifest.csv`](/d:/Project/Dyslexia_Detection_System/data/demo/audio_augmented_manifest.csv)	Audio-aug

3.2 Subfolders

Folder	Contents
[`data/demo/audio/`](/d:/Project/Dyslexia_Detection_System/data/demo/audio)	Demo audio assets
[`data/demo/handwriting/`](/d:/Project/Dyslexia_Detection_System/data/demo/handwriting)	Demo handwriting assets
[`data/demo/augmented/`](/d:/Project/Dyslexia_Detection_System/data/demo/augmented)	Augmented demo samples
[`data/demo/splits/`](/d:/Project/Dyslexia_Detection_System/data/demo/splits)	Train/validation/test splits
[`data/demo/audio_aug_splits/`](/d:/Project/Dyslexia_Detection_System/data/demo/audio_aug_splits)	Audio-focused splits

4. Collection Workspace

The collection folders support a real data-gathering workflow.

4.1 Main files

File	Role
[`data/collection/manifest.csv`](/d:/Project/Dyslexia_Detection_System/data/collection/manifest.csv)	Active collection manifest
[`data/collection/manifest_template.csv`](/d:/Project/Dyslexia_Detection_System/data/collection/manifest_template.csv)	Blank template for new collections

4.2 Subfolders

Folder	Contents
[`data/collection/raw/`](/d:/Project/Dyslexia_Detection_System/data/collection/raw)	Raw collected samples
[`data/collection/augmented/`](/d:/Project/Dyslexia_Detection_System/data/collection/augmented)	Augmented collection samples
[`data/collection/processed/`](/d:/Project/Dyslexia_Detection_System/data/collection/processed)	Cleaned and prepared samples
[`data/collection/splits/`](/d:/Project/Dyslexia_Detection_System/data/collection/splits)	Prepared train/validation/test splits
[`data/collection/docs/`](/d:/Project/Dyslexia_Detection_System/data/collection/docs)	Collection notes and supplementary data
[`data/collection/eye_tracking_smoke/`](/d:/Project/Dyslexia_Detection_System/data/collection/eye_tracking_smoke)	Eye-tracking smoke-test data

5. Benchmark Datasets

The benchmark folder is for more realistic or stricter evaluation.

5.1 Primary benchmark manifest

File
[`data/benchmarks/tough_confidence_manifest.csv`](/d:/Project/Dyslexia_Detection_System/data/benchmarks/tough_confidence_manifest.csv)

5.2 Hard split folders

Folder
[`data/benchmarks/hard_family_split/`](/d:/Project/Dyslexia_Detection_System/data/benchmarks/hard_family_split)
[`data/benchmarks/hard_family_split_strict/`](/d:/Project/Dyslexia_Detection_System/data/benchmarks/hard_family_split_strict)
[`data/benchmarks/hard_family_split_balanced/`](/d:/Project/Dyslexia_Detection_System/data/benchmarks/hard_family_split_balanced)
[`data/benchmarks/hard_family_split_realistic/`](/d:/Project/Dyslexia_Detection_System/data/benchmarks/hard_family_split_realistic)
[`data/benchmarks/hard_family_split_balanced_harder/`](/d:/Project/Dyslexia_Detection_System/data/benchmarks/hard_family_split_balanced_harder)

6. Biomarker Test Files

These CSVs are synthetic and are only intended for dashboard and UI testing.

File
[`data/biomarker_tests/english_biomarker_test.csv`](/d:/Project/Dyslexia_Detection_System/data/biomarker_tests/english_biomarker_test.csv)
[`data/biomarker_tests/bengali_biomarker_test.csv`](/d:/Project/Dyslexia_Detection_System/data/biomarker_tests/bengali_biomarker_test.csv)

7. Dataset Usage Notes

- Use demo data for quick local checks and UI validation.
- Use collection data only after anonymization and manifest validation.
- Use benchmark data for model selection and hard holdout sanity checks.

- Treat biomarker test CSVs as synthetic interface assets, not as training evidence.
- Keep all learner identities anonymized before training or reporting.

8. Related Docs

- docs/PROJECTDOCUMENTATION.md
(/d:/Project/DyslexiaDetectionSystem/docs/PROJECTDOCUMENTATION.md)
- docs/furtherresearch/experimentmatrix.md
(/d:/Project/DyslexiaDetectionSystem/docs/furtherresearch/experimentmatrix.md)
- docs/MODELCATALOG.md (/d:/Project/DyslexiaDetectionSystem/docs/MODELCATALOG.md)